When Does Observational Coherence Determine Probability?

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Abstract

When does observationally coherent data determine a genuine probability measure? We work with a directed system of Boolean algebras carrying a compatible family of finitely additive charges. Coherence and finite additivity are purely finitary conditions; the question is what further condition licenses the passage to σ-additivity.

Finitary coherence alone does not suffice: the finite–cofinite charge on $\mathbb{N}$ satisfies every structural condition yet assigns persistent mass to a sequence of events that vanishes. More generally, σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12]. Some condition beyond finitary coherence is therefore genuinely required.

We call this condition *collective exhaustion* (CE). A compatible charge family extends to a unique σ-additive probability measure if and only if it satisfies CE; equivalently, if and only if the purely finitely additive part in the Yosida–Hewitt decomposition vanishes at every level. Under a separation hypothesis (discriminability), CE is further equivalent to concentration of the Stone–space measure on realised states. Two constructions — Carathéodory and Stone — realise the extension.

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1 Introduction

A probability measure on a field $\mathcal{F} \subseteq 2^{\Omega}$ is a set function $P : \mathcal{F} \to [0, 1]$ satisfying [1, 10]:

$$\text{(positivity)} \qquad P(A) \geq 0; \tag{1}$$

$$\text{(normalisation)} \qquad P(\Omega) = 1; \tag{2}$$

$$\text{(countable additivity)} \qquad P\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} P(A_n) \quad (\text{disjoint } A_n,\ \bigcup A_n \in \mathcal{F}). \tag{3}$$

For a finitely additive P, (3) is equivalent to (e.g. [1], Theorem 2.1)

$$E_n \downarrow \varnothing \quad \Longrightarrow \quad P(E_n) \to 0 : \tag{4}$$

mass cannot persist along a vanishing sequence of events. In the Kolmogorov [10] and Carathéodory [2] extension theorems, this is postulated: σ-additive marginals are bundled into the input.

The directed structure is already present in the classical problem: compatible finite-dimensional distributions live on a directed family of coordinate projections. We isolate that structure — a directed system $(\iota, \leq)$ of event algebras $\mathcal{E}_i$ with surjective refinement maps

$$\pi_{ij} : O_j \to O_i \qquad (i \leq j), \qquad \pi_{ik} = \pi_{ij} \circ \pi_{jk} \qquad (i \leq j \leq k)$$

— and ask what the directed structure alone determines. Positivity and normalisation become local: each ℓ_i is a normalised finitely additive charge on $\mathcal{E}_i$. Compatibility replaces agreement of marginals:

$$\ell_j\big(\pi_{ij}^{-1}(A)\big) = \ell_i(A) \qquad (i \leq j,\ A \in \mathcal{E}_i). \tag{5}$$

These are finitary. Countable additivity (3) is not.

The finite–cofinite algebra on $\mathbb{N}$ shows how it fails. Set $\mu(A) = 0$ for finite A, $\mu(A) = 1$ for cofinite A. Then μ satisfies (1)–(5), yet

$$E_n = \{n, n{+}1, \ldots\} \downarrow \varnothing, \qquad \mu(E_n) = 1 \;\; \forall\, n :$$

mass persists because $\bigcap_n E_n = \varnothing$ is a countable fact no finite check can see. In general, σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12, 8, 5]. If σ-additivity is to be *recovered* rather than assumed, the condition must fall on the charges themselves.

We call this condition *collective exhaustion* (CE):

$$E_n \in \mathcal{E}_i, \quad E_n \downarrow \varnothing \quad \Longrightarrow \quad \exists\, j \geq i \text{ s.t. } \ell_j\big(\pi_{ij}^{-1}(E_n)\big) \to 0. \tag{6}$$

A compatible charge family extends to a unique σ-additive measure on $\sigma(\mathcal{C})$ if and only if it satisfies (6) (Theorem 6.1); under discriminability, this is equivalent to concentration of the Stone–space measure on realised states (Corollary 6.2). Two constructions realise the extension — one by Carathéodory (§3), one by Stone duality (§5).

2 Observational structure and compatible charges

Definition 2.1 (Observational structure)**.** An *observational structure* consists of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathsf{O}_i, \mathcal{O}_i)$, the *outcome space at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$, the *refinement map*,

satisfying $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$ for all $i \leq j \leq k$.

Example 2.2 (Successive observations of a dynamical system)**.** Let $T : X \to X$ and $h : X \to S$ with S finite. Set $\iota = (\mathbb{N}, \leq)$,

$$\mathsf{O}_L = S^L, \qquad \mathrm{eval}_L(x) = \big(h(x),\, h(Tx),\, \ldots,\, h(T^{L-1}x)\big),$$

and let $\pi_{ML} : S^L \to S^M$ drop the last $L - M$ entries. Passing from level L to $L + 1$ appends one observation:

$$\mathrm{eval}_{L+1}(x) = \big(\mathrm{eval}_L(x),\, h(T^L x)\big).$$

The canonical instantiation is $\Omega = S^{\mathbb{N}}$ and a compatible family $\{\ell_L\}$ is a consistent system of finite-dimensional distributions of $h, h \circ T, h \circ T^2, \ldots$

Definition 2.3 (Instantiation; query system)**.** An *instantiation* is a set Ω with *evaluation maps* $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ satisfying

$$\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i \qquad (i \leq j).$$

A *query system* is an observational structure together with an instantiation.

Every observational structure admits a canonical instantiation: the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathsf{O}_i = \Big\{\omega \in \prod_{i \in \iota} \mathsf{O}_i \;\Big|\; \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j\Big\},$$

with $\mathrm{eval}_i(\omega) = \omega_i$. In Example 2.2, $\Omega = S^{\mathbb{N}}$ is the space of complete observed trajectories: each $\omega \in \Omega$ is a full record

$$\omega = \big(h(x),\, h(Tx),\, h(T^2x),\, \ldots\big)$$

and $\mathrm{eval}_L(\omega) = (\omega_1, \ldots, \omega_L)$ extracts the first L entries.

Definition 2.4 (Common coarsenings)**.** The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$.

Definition 2.5 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{O}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder family* is $\mathcal{C} = \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{O}_i\}$, and the *observable σ-algebra* is $\sigma(\mathcal{C})$.

In Example 2.2, $\mathrm{Cyl}(L, A)$ is the set of trajectories whose first L observations fall in $A \subseteq S^L$:

$$\mathrm{Cyl}(L, A) = \big\{\omega \in S^{\mathbb{N}} : (\omega_1, \ldots, \omega_L) \in A\big\}.$$

Lemma 2.6 (Cylinder π-system)**.** *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. The coherence condition gives $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ for $i \leq j$: the same subset of Ω is a cylinder at any finer level. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

For each $i \in \iota$, let B_i denote the Boolean algebra of level-i cylinders $\{\mathrm{Cyl}(i, A) : A \in \mathcal{O}_i\}$; then $\mathcal{C} = \bigcup_{i \in \iota} B_i$. For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

Definition 2.7 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i \in \iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \leq j \text{ and all } E \in B_i.$$

In Example 2.2, compatibility says that the probability assigned to an event described by the first M observations does not change when the window is extended to $L > M$:

$$\ell_L\big(\{s \in S^L : (s_1, \ldots, s_M) \in A\}\big) = \ell_M(A) \qquad (A \subseteq S^M).$$

Compatibility ensures a well-defined charge on $\mathcal{C}$ via $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$.

Definition 2.8 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathsf{O}_i$ is surjective for every $i \in \iota$.

Lemma 2.9 (Surjectivity and injectivity from Surjective Evaluation)**.** *If the query system satisfies Surjective Evaluation, then for every $i \leq j$:*

(i) the refinement map $\pi_{ij} : \mathsf{O}_j \to \mathsf{O}_i$ is surjective;

(ii) the connecting map $\varphi_{ij} : B_i \to B_j$, defined by $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$, is an injective Boolean algebra homomorphism.

Proof. For (i): given $y \in \mathsf{O}_i$, surjectivity of eval_i yields $\omega \in \Omega$ with $\mathrm{eval}_i(\omega) = y$. Then $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega) = y$ by the coherence condition, so π_{ij} is surjective.

For (ii): φ_{ij} is a Boolean homomorphism by construction. If $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \varphi_{ij}(\mathrm{Cyl}(i, A'))$, then $\pi_{ij}^{-1}(A) = \pi_{ij}^{-1}(A')$ in O_j. Applying π_{ij} and using surjectivity gives $A = A'$, so φ_{ij} is injective. □

3 Direct extension from compatible probability marginals

Kolmogorov's extension theorem [10] takes as input compatible σ-additive marginals

$$(\pi_{F'F})_{\#}\nu_{F'} = \nu_F \qquad (F \subseteq F' \subset T,\ |F'| < \infty), \qquad \nu_F \in \mathcal{P}(\mathbb{R}^F),$$

and extends to a unique measure on $\mathbb{R}^T$ via tightness. Compatibility (5) and σ-additivity of each ν_F are bundled into the hypothesis. Retaining both, what replaces tightness? An order-theoretic condition on ι:

Definition 3.1 (Sequential common refinements)**.** The index set $(\iota, \leq)$ admits *sequential common refinements* if every countable $\{i_n\}_{n\geq 1} \subset \iota$ has an upper bound in ι.

Any measure that agrees on cylinders is determined on $\sigma(\mathcal{C})$:

Lemma 3.2 (Observational determination)**.** *If $(\iota, \leq)$ admits common coarsenings and P, P' are probability measures on $(\Omega, \sigma(\mathcal{C}))$ with $(\mathrm{eval}_i)_{\#}P = (\mathrm{eval}_i)_{\#}P'$ for all i, then $P = P'$.*

Proof. P and P' agree on $\mathcal{C}$; apply the π–λ theorem to $\sigma(\mathcal{C})$. □

Theorem 3.3 (Carathéodory observational extension)**.** *Suppose $(\iota, \leq)$ admits common coarsenings and sequential common refinements, the query system satisfies Surjective Evaluation, and $\{\nu_i\}$ is a compatible family of probability measures on $\{(\mathsf{O}_i, \mathcal{O}_i)\}$. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ with*

$$(\mathrm{eval}_i)_{\#}P = \nu_i \qquad \textit{for every } i \in \iota.$$

Proof. Content. The assignment $\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A)$ is well-defined by compatibility: if $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$, pass to a common upper bound $k \geq i, j$ and use

$$\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B).$$

Finite additivity of each ν_k gives finite additivity of μ_0.

σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(i_n, A_n)$. Sequential common refinements yield a single k with $k \geq i$ and $k \geq i_n$ for all n. Compressing to level k:

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k\big(\pi_{ik}^{-1}(B)\big) \leq \sum_{n=1}^{\infty} \nu_k\big(\pi_{i_n k}^{-1}(A_n)\big) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Extension. The cylinder family $\mathcal{C}$ is a π-system (Lemma 2.6) closed under relative differences — for $E = \mathrm{Cyl}(i, A)$ and $F = \mathrm{Cyl}(j, B)$,

$$E \setminus F = \mathrm{Cyl}\big(k,\ \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)\big) \in \mathcal{C}$$

— so $\mathcal{C}$ is a semiring. Carathéodory's theorem (Bogachev [2], I.1.11.9) extends μ_0 to a measure P on $\sigma(\mathcal{C})$.

Uniqueness. Any two extensions agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by the π–λ theorem. □

4 Collective Exhaustion

Theorem 3.3 assumes σ-additive marginals. What if only finitely additive charges are given? At a single algebra, a finitely additive charge extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if it satisfies (4) (Halmos [7]). But verifying $\bigcap_n E_n = \varnothing$ requires access to $\sigma(\mathcal{E})$, which $\mathcal{E}$ alone cannot see. In a directed system, a finer level might witness what a coarser level cannot — but the following example shows it need not:

Example 4.1 (Finite–cofinite obstruction)**.** On $\mathbb{N}$, set $\mu_0(A) = 0$ for finite A, $\mu_0(A) = 1$ for cofinite A. Then μ_0 is normalised and finitely additive, but $E_n = \{n, n+1, \ldots\} \downarrow \varnothing$ with $\mu_0(E_n) = 1$ for all n.

The condition that rules out this failure is:

Definition 4.2 (Collectively exhaustive family)**.** The family is *collectively exhaustive* if for every $i \in \mathcal{I}$ and every $E_n \downarrow \varnothing$ in $\mathcal{E}_i$,

$$\exists\, j \geq i \text{ such that } \ell_j\big(\pi_{ij}^{-1}(E_n)\big) \to 0.$$

No first-order condition can force CE: σ-additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [12, 11]. It is, however, the exact condition needed:

Theorem 4.3 (CE Characterisation)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every i.

Proof. (i) ⇒ (ii). Fix i and $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. Compatibility gives $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$, so ℓ_i satisfies (4) and extends.

(ii) ⇒ (i). If ℓ_i extends to μ_i, then $\ell_i(E_n) = \mu_i(E_n) \to 0$ by σ-additivity. Take $j = i$. □

The Yosida–Hewitt decomposition [15] makes the content precise:

$$\ell = \ell_c + \ell_p, \qquad \ell_c \text{ countably additive}, \quad \ell_p \text{ purely finitely additive}.$$

CE is $\ell_p = 0$.

5 Stone realization from finitely additive data

No first-order condition forces CE (§4). But σ-additivity need not come from the charges. The cylinder algebra $\mathcal{C}$ is a Boolean algebra; its Stone space $\mathrm{St}(\mathcal{C})$ is compact; and any finitely additive charge on a compact Boolean algebra extends to a σ-additive Baire measure — unconditionally. The resulting measure lives on $\mathrm{St}(\mathcal{C})$, not on Ω:

$$\{\ell_i\} \xrightarrow{\text{Stone}} \hat{\mu} \text{ on } \mathrm{St}(\mathcal{C}) \xrightarrow{\text{descend}} P \text{ on } (\Omega, \sigma(\mathcal{C})).$$

The first arrow is free. The second is not. Each $\omega \in \Omega$ determines a principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$, but $\mathrm{St}(\mathcal{C})$ also contains non-principal ultrafilters — ideal points corresponding to no realised state — and $\hat{\mu}$ may assign them positive mass. The question is whether

$$\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1.$$

Proposition 5.1 (Stone duality for the direct limit)**.** $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.

Proof. The connecting maps φ_{ij} are injective (Lemma 2.9(ii)); St sends colimits of Boolean algebras to limits of compact Hausdorff spaces [9, II.4]. □

Proposition 5.2 (Inverse limit measure)**.** *A compatible family of normalised charges $\{\mu_i\}$ on $\{B_i\}$ extends to a unique Baire probability measure $\hat{\mu}$ on* $\mathrm{St}(\mathcal{C})$*.*

Proof. Each μ_i extends to a regular Borel measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$ (compact, totally disconnected; clopens form a base [3, 7]). Compatibility translates to $(\pi^i_j)_\# \hat{\mu}_j = \hat{\mu}_i$. Choksi [4, Theorem 1] extends the compatible family to the inverse limit. □

Definition 5.3 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ separating them.

Write $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ and $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$. Discriminability makes pure injective.

Proposition 5.4 (Structural identification)**.** *Under Discriminability,*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}).$$

Proof. ($\supseteq$): $\mathrm{pure}^{-1}([E]) = E$ for each cylinder E, and $[E]$ is clopen hence Baire. ($\subseteq$): clopens of $\mathrm{St}(\mathcal{C})$ are exactly $[E]$ for $E \in \mathcal{C}$ [9, I.3.2], so pure^{-1} maps Baire sets into $\sigma(\mathcal{C})$. Injectivity of pure ensures the pullback does not conflate events. ☐

Proposition 5.5 (Support condition)**.** *If μ satisfies CE, then $\hat{\mu}$ is supported on* $\mathrm{pure}(\Omega)$.

Proof. By Theorem 4.3, CE gives $\mu_p = 0$, so μ is σ-additive. The corresponding Baire measure on $\mathrm{St}(\mathcal{C})$ assigns full mass to ultrafilters closed under countable intersection [13, Ch. 10]. Each principal ultrafilter $\mathrm{pure}(\omega)$ is closed under countable intersection, so $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$. ☐

Theorem 5.6 (Stone observational extension)**.** *Under Surjective Evaluation, Discriminability, common coarsenings, and CE, a compatible family of normalised charges extends to a unique $P \in \mathcal{P}(\Omega, \sigma(\mathcal{C}))$ with*

$$P(\mathrm{Cyl}(i, A)) = \mu_i(\mathrm{Cyl}(i, A)) \qquad (i \in \iota,\ A \in \mathcal{O}_i).$$

Proof. Proposition 5.2 gives $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$. Proposition 5.5 gives $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$. Define

$$P(A) = \hat{\mu}\big(\mathrm{pure}(A)\big) \qquad (A \in \sigma(\mathcal{C}));$$

well-definedness follows from Proposition 5.4 and injectivity of pure. Uniqueness by Lemma 3.2. ☐

6 Synthesis

Kolmogorov's hypothesis — compatible σ-additive marginals — decomposes into a finitary part (compatibility (5)) and an infinitary part (countable additivity (3)). The three preceding sections show these have equivalent formulations at each layer of the construction:

Theorem 6.1 (Main equivalence)**.** *Let $\{\ell_i\}$ be a compatible family of normalised finitely additive charges on a directed system. Write $\ell_i = \ell_{i,c} + \ell_{i,p}$ for the Yosida–Hewitt decomposition [15]. The following are equivalent:*

(i) Each ℓ_i is σ-additive.

(ii) The family is collectively exhaustive.

(iii) $\ell_{i,p} = 0$ for every i.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 4.3. (ii) $\Leftrightarrow$ (iii): CE holds iff each ℓ_i extends σ-additively; the Yosida–Hewitt decomposition identifies this with $\ell_{i,p} = 0$. ☐

Corollary 6.2. *Under Discriminability (Definition 5.3),* (i)–(iii) *are each equivalent to*

$$\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1,$$

where $\hat{\mu}$ is the Baire measure on $\mathrm{St}(\mathcal{C})$.

Proof. Proposition 5.5. ☐

When any of these holds and $(\iota, \leq)$ admits sequential common refinements, the Carathéodory and Stone extensions (Theorems 3.3 and 5.6) yield the same measure by Lemma 3.2.

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